McGill University
Department of Mathematics and Statistics
MATH 784 Advanced Topics in Probability

Student Name: Wen, Zehai Student ID# 260956530

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Degree Program and Year: PhD 2 Term: Fall Winter

Topics to be covered: (Outline the Main reading material.)

Topic Title: High-Dimensional Probability

Prerequisites: MATH 356 (Probability) or equivalent, and familiarity with norms, vector spaces, and convexity.

References (no required textbook): Vershynin, High-Dimensional Probability (free online); Wainwright, High-Dimensional Statistics

About the course. This course develops the theory of high-dimensional probability: random vectors and matrices, and the mathematics of how they transform under norms, eigenvalue maps, and other (often convex) operations. The subject reveals fundamental geometric features of normed spaces and convex sets in high dimensions, and the material has substantial applications to topics in statistics and machine learning — including covariance estimation, random matrix theory, and the analysis of high-dimensional estimators

Topics will include:
- Norms and eigenvalues of random matrices (operator norm bounds via nets, covariance estimation, the Marchenko–Pastur law for sample covariance spectra, smallest & largest singular values)
- Gaussian processes and their extremes (Sudakov–Fernique and Gordon comparison inequalities, expected suprema, Gaussian width, Lipschitz and log-Sobolev concentration)
- Chaining (Dudley's entropy integral, Talagrand's generic chaining, with applications to nonparametric regression and empirical process bounds)
- Selected applications (Dvoretzky–Milman, further statistical applications)

Method of Evaluation: Homework (50%) + Oral Exam (50%)

Student Signature and Date:

Instructor Name: Elliot Paquette Signature and Date:

For Undergraduate Students - Chief Advisor Approval:

Professor Djivede Kelome Signature and Date:

For Graduate Students – Graduate Program Director Approval

Professor Brent Pym Signature and Date:

Please return form with all 3 signature via email to the Undergraduate Program Coordinator, Angela White at angela.white@mcgill.ca

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